Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3481-WE01

Title: (BEng/MEng)

Power Electronic Devices 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

A made-up semiconductor, carbonium, has the band structure shown in the dispersion diagram in **Fig. Q1**. The conduction band (CB) has two local minima points, commonly known as valleys. The central one is marked with Γ, at $k = 0$. The second one is marked with Λ and has a symmetric point at $\Lambda' = -\Lambda$, which will not be considered in this question. The valence band (VB) has a single maximum point at Γ. E_V is the reference point for the system, so that $E_V = 0$.

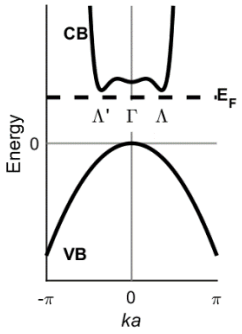


Figure Q1

The dispersion relation for the conduction band is given by:

$$CB(k) = E_0(a^6k^6 - 2.25a^4k^4 + a^2k^2) + E_1$$

Where $E_0 = 0.6$ eV, $E_1 = 1.4$ eV, and $a = 8$ Å.

a. i. Without calculation and based on **Fig. Q1**, is carbonium a direct or indirect band gap semiconductor? Explain your answer.
ii. Calculate the k location of the Λ valley in the conduction band.
iii. Calculate the energy differences between E_V and the two valleys. Term them $E_{\Gamma-\Gamma}$ and $E_{\Gamma-\Lambda}$. Give your answer in eV.

[20%]

b. i. Derive a general expression for the effective mass in the conduction band, as a function of k.
ii. Calculate the effective masses for electrons in the two valleys, terming them m^*_Γ and m^*_Λ. Give your answer in m_e units. If you did not find an answer to **part a.iii**, you may use $k_\Lambda = 1.2/a$.

[28%]

For conduction purposes the two valleys act independently, each having characteristic mobilities, effective densities of states and electron densities. In carbonium, both valleys have the same mean time for scattering, $\tau = 2$ ps.

c. Calculate the conductivity of carbonium, if the Fermi level is located 0.1 eV below the Λ valley minimum. Explain your steps and any approximation you make. If you did not find an answer to **part b.ii.** you may use $m^*_\Gamma = 0.3\, m_e$ and $m^*_\Lambda = 0.05\, m_e$.

[28%]

We now illuminate the sample with a wavelength that corresponds to the $E_{\Gamma-\Gamma}$ transition, generating an excess of electrons Δn.

d. i. Write an expression for the change in conductivity as a function of Δn, before any relaxation process (valley to valley transition of electrons) can occur.
ii. If we define steady state relaxation as a condition in which the quasi-Fermi level for electrons is uniform across the k space, what would be the fraction of Δn electrons left in the Γ valley in steady state relaxation? If you did not find an answer to **part b.ii.** you may use $m^*_\Gamma = 0.3\, m_e$ and $m^*_\Lambda = 0.05\, m_e$.

[24%]

Continued

Question 2

A cuboid piece of GaAs ($n_i = 2 \times 10^6$ cm^{-3}) that is doped only with donors ($N_D = 10^{18}$ cm^{-3}) is biased with a DC power supply sourcing 1 V in the configuration shown in **Fig. Q2(a)**. The measured current is 1.3 mA. The length of the sample is $L = 1$ μm and its width and depth are both 0.1 μm.

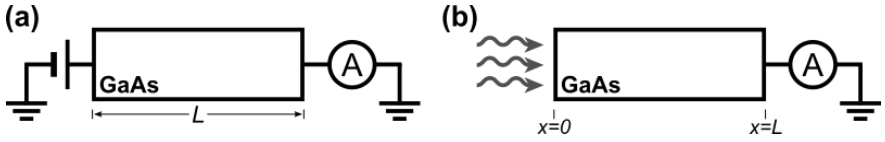


Figure Q2

a. Calculate the conductivity of the GaAs piece.

[20%]

The power source at $x = 0$ is now disconnected, but the electrode at $x = L$ is still grounded, as shown schematically in **Fig. Q2(b)**. The surface at $x = 0$ is illuminated with a constant source producing a steady state excess of carriers, $\delta p = 10^{15}$ cm^{-3}.

b. i. Show that in this system, the diffusion length for minority carriers is significantly longer than the length of the semiconductor, if it is known that the mobility ratio is $\mu_n/\mu_p = 20$, and the lifetime for holes is $\tau_p = 0.1$ μs. If you did not find an answer to **part a**, you may use $\sigma = 10^3$ Ω^{-1}cm^{-1}

ii. Calculate the full profile of the excess minority carriers across the piece, assuming that on the grounded side, the electrode forces zero excess of carriers ($\Delta p(L) = 0$). To solve your equation, arrange it in the form $y'' = a^2 y$, for which the solution is:

$$y = A\exp\big(a(x - x_0)\big) + B\exp\big(-a(x - x_0)\big)$$

Give your answer as a function of x, using L and δp as parameters. Show that the solution is proportional to $\sinh[(L - x)/L_p]$.

iii. When $\theta < 0.1$, we can use the approximation $\sinh(\theta) \approx \theta$. Derive a linear expression for $\Delta p(x)$ using this approximation. What will be the net current measured at the electrode at $x = L$?

[60%]

c. The excess of photogenerated carriers at $x = 0$ can be translated to a voltage on that electrode, known as a photovoltage. The photovoltage, V_{pv}, is given by the split between the quasi-Fermi levels. Calculate V_{pv} at $x = 0$ for the case where $\delta p = 10^{15}$ cm^{-3}.

[20%]

Continued

Question 3

a. For an abrupt junction p-i-n diode manufactured from silicon, calculate the specific on-state resistance of the device if the intrinsic layer has the following properties:
   Epilayer thickness 10 μm.
   Donor Concentration 1×10^{14} cm^{-3}.

[16%]

b. Calculate the voltage drop of the diode if it operates at a current density of 250 Acm^{-2}, stating any assumptions made.

[16%]

c. A silicon carbide diode is manufactured from an epilayer of identical dopant concentration to the silicon diode in **part a**. What is the thickness of the epilayer if the power loss during conduction (i.e. under forward bias) in the two diodes is equal?

[40%]

d. Calculate the breakdown voltage of the silicon carbide diode under the assumption that it behaves as an abrupt junction. Comment on the practicality of the calculated value in a real device.

[28%]

Question 4

a. Sketch a schematic cross-section for a UMOS power MOSFET device and create a process flow suitable for its manufacture

[40%]

b. Calculate the flatband voltage, V_{FB}, of a MOS capacitor manufactured from aluminium (work function 4.08 eV), silicon carbide (electron affinity 4.75 eV), and silicon dioxide as the dielectric, if the fixed oxide charge is -1.25×10^{-7} C/cm^2. The acceptor concentration in the silicon carbide is 1×10^{15} cm^{-3} and the gate dielectric is SiO_2 of thickness 20 nm. The gate width of the transistor is 200 μm and the gate length 2 μm.

[24%]

c. The capacitor is used to form the gate of a MOSFET. Determine the threshold voltage of the MOSFET if the dopant concentration in the drift region is 1×10^{15} cm^{-3}.

[20%]

d. Calculate the saturation current in the MOSFET if the gate voltage is set to 20 V.

[16%]

END